

KOUTIALA, Mali

WMO: 612930

Lat: 12.383N

Lon: 5.467W

Elev: 367

StdP: 96.99

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 16.4	(c) 17.6	(d) -5.2	(e) 2.5	(f) 29.5	(g) -3.3	(h) 3.0	(i) 29.1	(j) 6.5	(k) 27.7	(l) 5.9	(m) 28.0	(n) 1.1	(o) 0	(p) 0.366

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 4	(b) 10.8	(c) 40.1	(d) 21.6	(e) 39.3	(f) 21.8	(g) 38.5	(h) 21.6	(i) 26.7	(j) 33.1	(k) 26.2	(l) 32.5	(m) 25.9	(n) 32.2	(o) 2.7	(p) 40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.3	21.4	29.2	24.9	20.8	28.7	24.5	20.4	28.5	86.1	33.2	84.0	32.5	82.4	32.1	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.5	(b) 5.8	(c) 5.1	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 6.5	(b) 5.8	(c) 5.1	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.6	25.0	28.4	31.7	33.0	32.2	29.6	27.3	26.5	27.2	29.2	28.1	25.7
	DBStd	3.05	1.91	2.52	2.08	1.78	2.01	2.03	1.59	1.31	1.44	1.40	1.41	1.77
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6806	466	514	671	690	687	588	537	512	517	595	542	487
	CDD18.3	3764	208	281	413	440	429	338	278	253	267	337	292	228
	CDH23.3	47746	2229	3641	6262	6861	6357	4408	2889	2247	2601	4088	3628	2536
	CDH26.7	26224	1069	2096	4007	4507	3974	2307	1133	713	1021	2082	2023	1292
Wind	WSAvg	2.4	2.2	2.4	2.2	2.4	3.2	3.2	2.9	2.4	2.0	2.1	1.7	2.0
Precipitation	PrecAvg	890	1	1	4	21	68	124	212	245	160	57	5	1
	PrecMax	1343	10	27	18	85	158	244	344	526	347	342	42	42
	PrecMin	515	0	0	0	0	10	26	84	0	44	7	0	0
	PrecStd	163	2	4	5	21	31	45	60	87	65	59	9	6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.3	39.7	40.6	41.1	40.6	38.6	35.5	32.8	34.4	37.0	37.0	36.1
		MCWB	17.7	20.3	20.4	21.5	22.4	24.8	24.9	25.4	25.7	22.2	20.2	18.7
	2%	DB	34.6	38.2	39.7	40.2	39.5	36.9	33.7	31.9	33.2	36.1	36.1	35.0
		MCWB	17.3	19.2	20.3	21.8	23.5	24.8	25.1	25.5	25.6	22.8	20.0	18.2
	5%	DB	33.3	37.1	38.9	39.3	38.5	35.7	32.5	31.1	32.3	35.2	35.4	34.0
		MCWB	16.6	18.7	20.1	22.0	24.2	24.9	25.0	25.3	25.5	23.6	19.8	17.9
	10%	DB	32.2	35.9	38.1	38.5	37.4	34.5	31.5	30.2	31.4	34.3	34.7	32.9
		MCWB	16.3	18.0	19.5	22.1	24.4	25.0	24.9	25.0	25.3	24.0	19.7	17.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.5	23.8	25.1	26.7	27.1	27.1	26.2	27.0	27.0	27.3	25.4	21.7
		MCDB	30.6	35.5	36.1	36.4	36.1	34.0	31.7	30.6	31.8	32.9	32.9	32.9
	2%	WB	19.5	22.3	24.1	25.6	26.4	26.4	25.7	26.1	26.4	26.2	24.0	20.3
		MCDB	29.2	33.6	35.1	35.0	34.9	33.2	31.0	30.2	31.3	32.2	31.6	31.6
	5%	WB	18.0	20.9	23.2	25.0	25.9	25.9	25.4	25.7	26.0	25.7	23.2	19.1
		MCDB	30.6	32.6	34.5	34.6	34.1	32.8	30.5	29.6	30.9	31.7	30.5	31.5
	10%	WB	17.0	19.2	22.4	24.5	25.5	25.5	25.1	25.2	25.6	25.2	22.1	18.0
		MCDB	30.3	33.5	33.5	34.1	33.7	32.2	30.1	29.1	30.2	31.2	29.7	31.0
Mean Daily Temperature Range	5% DB	MDBR	14.1	14.0	13.0	10.8	10.2	9.1	7.9	7.1	8.3	10.6	14.3	14.9
		MCDBR	16.1	15.4	13.8	12.1	11.2	10.5	9.2	8.3	9.6	11.9	15.0	16.3
		MCWBR	5.3	5.2	4.8	4.3	3.2	2.9	2.9	2.9	3.4	3.4	4.3	4.9
	5% WB	MCDBR	14.8	12.4	11.0	10.4	10.4	9.3	8.4	7.8	9.1	10.2	12.0	15.4
		MCWBR	6.0	5.3	4.9	3.7	3.2	3.3	3.2	3.2	3.6	3.6	4.4	5.6
Clear-Sky Solar Irradiance	taub	0.524	0.582	0.715	0.710	0.674	0.641	0.571	0.537	0.536	0.570	0.527	0.500	
	taud	1.880	1.743	1.517	1.605	1.662	1.757	1.943	2.036	2.022	1.909	1.910	1.942	
	Ebn at Noon	777	750	665	666	677	691	745	781	787	750	769	786	
	Edh at Noon	197	234	299	272	252	226	189	174	177	195	189	181	
All-Sky Solar Radiation	RadAvg	5.60	5.99	6.30	6.31	6.21	5.88	5.61	5.37	5.66	5.93	5.68	5.46	
	RadStd	0.31	0.30	0.30	0.30	0.17	0.16	0.15	0.23	0.27	0.15	0.17	0.20	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page